Specialist Physical AI Need Not Be Opaque
A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control and exact weight analysis

Anonymous Author(s)
Affiliation
Address
email

Abstract

Tensor decomposition turns a large table of learned interactions into simpler low-rank factors that can be ranked and inspected. Conventional robot policies contain operations that block this direct weight-level view. We introduce χ-VLA, a 20.1M-parameter vision-language-action policy built only from bilinear, linear, and rational learned operations. Its weights can therefore be reorganized into explicit interaction tensors rather than treated only as an opaque checkpoint. A mechanical audit and numerical reconstructions verify that the deployed policy preserves this form. Trained from scratch on LIBERO-Object, convolutional χ-VLA reaches $93.7\% \pm 0.8\%$ over three seeds under the published protocol. In a separate ViT instantiation, we reconstruct every attention module in three checkpoints with worst relative error 2.56×10^{-7}, and exact weight-derived subspaces beat equal-rank random controls in all 36 primary block-head tests. The result is a capable and structurally inspectable specialist family with exact access to trained attention coefficients. It does not yet provide an exact end-to-end decomposition of the full policy.

1 Introduction

Modern robot policies can achieve high success while giving little indication of why an action was chosen. Their checkpoints contain millions of weights, but those weights do not directly show which image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated activations, yet a correlation is not automatically a mechanism or a safe editing target.

A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction table defined by its weights. Its decomposed factors then provide candidate control mechanisms that can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully tensor-decomposable policy.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

We study the narrower but testable question:

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Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?

We test capability and structural analysis in two fully rational encoder instantiations. The convolutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact attention audit.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r\times(d+1)}$ and $D \in \mathbb{R}^{d\times r}$,

$$\mathrm{BFFN}(x) = D\left[(L\bar{x}) \odot (R\bar{x})\right], \qquad y_o = \sum_{i,j}\left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j}\right)\bar{x}_i \bar{x}_j. \tag{1}$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C\left[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top)/s\right] V. \tag{2}$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with interaction coefficients determined by the weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z+\epsilon}$:

$$\mathrm{RNorm}(x) = x\, r\left(\frac{1}{d}\sum_{j=1}^{d} x_j^2\right). \tag{3}$$

Here rational means a ratio of polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D. Updating this pair preserves input-specific rescaling without leaving the rational tensor representation.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has $20{,}137{,}352$ learned parameters.

Training uses $66{,}984$ demonstration frames, $40{,}000$ AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real $189/200$ checkpoint at runtime and reconstruct representative operations from saved weights.

The normalization reconstruction has maximum absolute error 1.64×10^{-6}. A bilinear FFN reconstruction has median relative error 1.40×10^{-7}. These residuals are bounded by the deployed module's float32 statistic. The audit certifies operator membership and local reconstruction, as summarized in Appendix Figure 3. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and $1{,}500$ rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

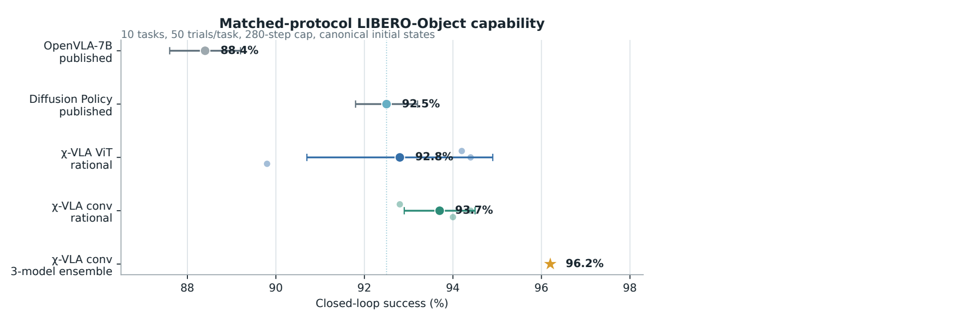

Figure 1: **Protocol-aligned LIBERO-Object capability.** Small dots are χ-VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ-VLA, rational conv	20.1M	3	$\mathbf{93.7\% \pm 0.8\%}$
χ-VLA, conv ensemble	3×20.1M	$3 \to 1$	$\mathbf{96.2\%}$
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high specialist capability is reproduced across three independently trained checkpoints. These are historical Modal aggregate records. A locked replay is producing immutable episode-level outcomes.

Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15}. The internal core identity agrees to 1.24×10^{-14}.

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16}. We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5}. This residual comes from the deployed implementation's intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors rank important directions without constructing the full table. It matches brute-force materialization to 2.13×10^{-14} at toy scale.

4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached activations are used only afterward to score how well it preserves the layer's computation.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033

and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ-nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The benchmark is one specialist suite, so it does not establish broad robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce one compact symbolic contraction of the full policy. The capability result uses convolutional checkpoints, while the exact analysis uses separate ViT checkpoints. Evaluation uses fixed task instructions and does not establish robust instruction grounding. Within that scope, rational restrictions support strong LIBERO-Object control and exact trained-weight certificates. We do not claim a direct coefficient-edit interface.

Reproducibility. The final artifact will provide code, configurations, checkpoint hashes, exact certificates, and per-episode logs. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A source-level audit compared all $271{,}996$ cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized images matched exactly for every frame. Canonical-state checks fail loudly when requested states are unavailable.

References

[1] M. J. Kim et al. *OpenVLA: An Open-Source Vision-Language-Action Model.* CoRL, 2024.

[2] M. J. Kim, C. Finn, and P. Liang. *Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success.* 2025.

[3] A. Rigg, T. Dooms, M. T. Pearce, J. Oramas, and L. Sharkey. *Bilinear MLPs Enable Weight-Based Mechanistic Interpretability.* ICLR, 2025.

[4] T. Dooms, W. Gauderis, G. A. Wiggins, and J. Oramas. *Compositionality Unlocks Deep Interpretable Models.* 2025.

[5] F. Oozeer, Y. Erisken, A. Rigg, J. Oramas, and L. Sharkey. *Bilinear Convolution Decomposition for Causal RL Interpretability.* 2024.

[6] B. Häon, K. Stocking, I. Chuang, and C. Tomlin. *Mechanistic Interpretability for Steering Vision-Language-Action Models.* 2025.

[7] S. Marks et al. *Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models.* ICLR, 2025.

[8] M. Kekić et al. *Learning Nonlinear Causal Reductions to Explain Reinforcement Learning Policies.* ICLR, 2026.

[9] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization.* 2025.

[10] Y. Fang et al. *When Vision Overrides Language: Evaluating and Mitigating Counterfactual Failures in VLAs.* 2026.

[11] C. Kim et al. *LIBERO-Para: A Diagnostic Benchmark and Metrics for Paraphrase Robustness in VLA Models.* 2026.

[12] T. Kolda and B. Bader. *Tensor Decompositions and Applications.* SIAM Review, 2009.

[13] E. Stoudenmire and D. Schwab. *Supervised Learning with Tensor Networks.* NeurIPS, 2016.

[14] G. Chrysos et al. *Deep Polynomial Neural Networks.* CVPR and TPAMI, 2020 to 2021.

A Supporting results and decisions

A.1 Per-task ensemble behavior

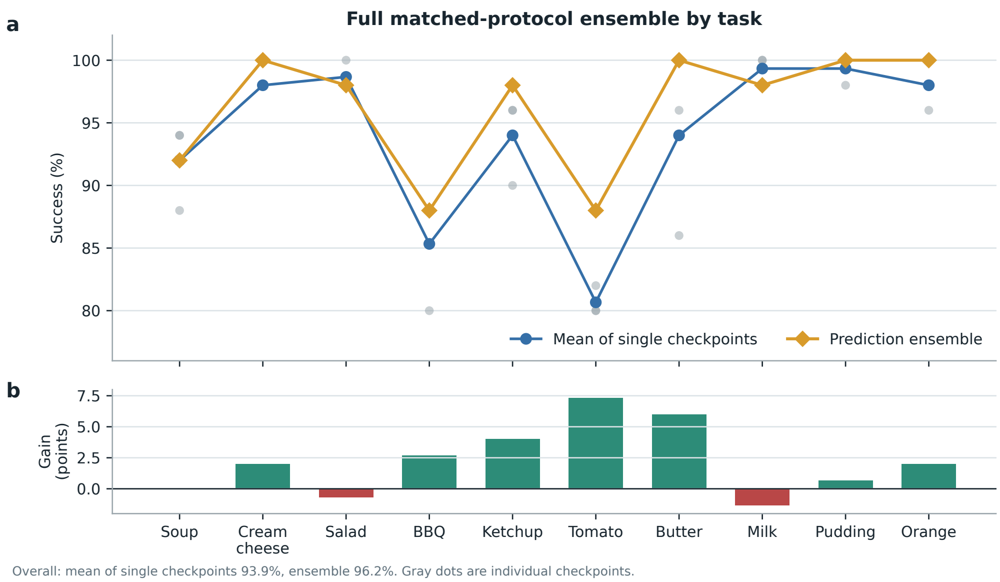

Figure 2: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

A.2 Expanded attention screen

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, $83/96$ block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only $54/96$ and $60/96$ pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.3 Numerical scope of rational conversion

The deployed Padé computation is exactly rational even though it approximates RMS normalization. Final certification must report its polynomial degrees, coefficients, denominator margin, observed activation range, tail error, action drift, and runtime overhead. Operator membership alone does not establish practical numerical safety under distribution shift.

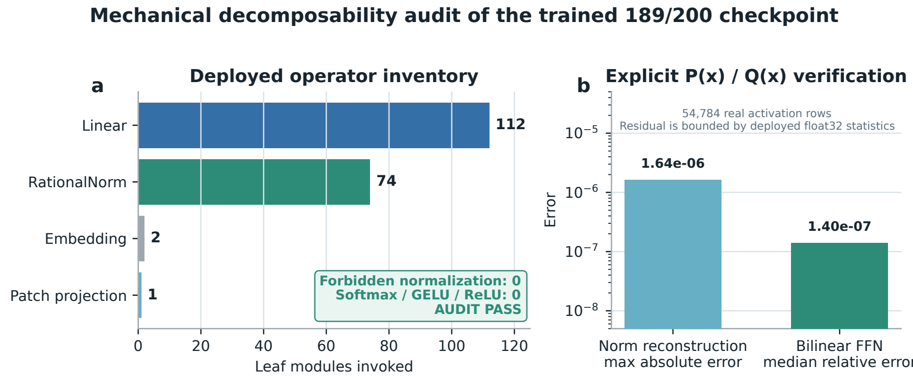

Figure 3: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over $54{,}784$ real activation rows.